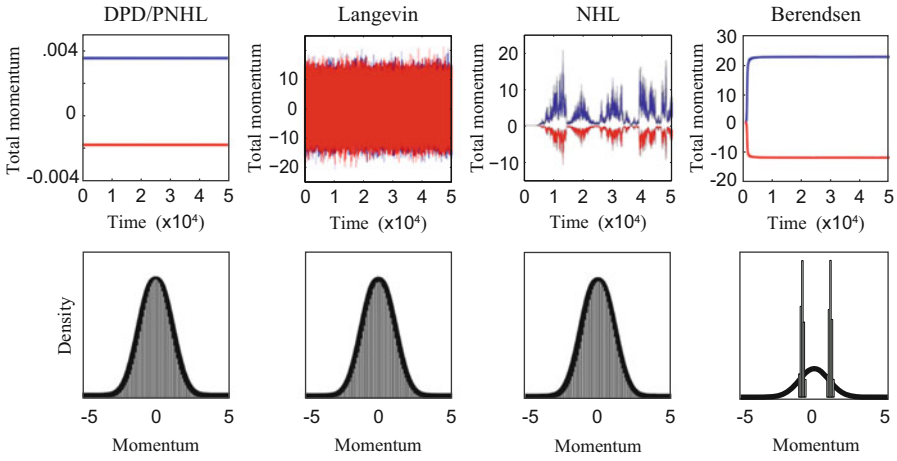


Fig. 8.8 We compare the time evolution of the system's total momentum in the x and y components (*top row*, marked *blue* and *red* respectively), and momentum distributions (*bottom row*) for various methods. In the *lower row* of figures, the *dark*, *solid curve* represents the exact normal density, while the superimposed histograms are constructed from the simulation data. From *left to right*: the PNHL-N thermostat, Langevin dynamics, Nosé-Hoover-Langevin (NHL) dynamics and the Berensen device. The same initial conditions were used with small values of total momenta. The Berendsen device shows a rapid and systematic increase in the momenta, NHL and Langevin exhibit different behavior, obtaining large fluctuating values, whereas the PNHL-N thermostat preserves the total momenta exactly. Standard DPD results are identical to those of PNHL in this respect

for some decay constant $\gamma_{\mathrm{T}} > 0$. The dynamics will not necessarily sample canonically, but the method will ensure that the average temperature is well-controlled (and is hence sometimes useful for equilibration purposes). We implement the version of the scheme given in [186], which completes a step of the velocity Verlet scheme followed by a scaling of the momenta:

$$\boldsymbol{p} \leftarrow \sqrt{1 + \gamma_{\mathrm{T}} h \left(\frac{\mathrm{N_d} k_{\mathrm{B}} \mathrm{T}}{\|\boldsymbol{p}\|^2} - 1 \right)} \boldsymbol{p},$$

for the given timestep h.

We use parameters $\gamma = 4.5$, $\mu = 1$ and $\gamma_{\mathrm{T}} = 1$ for the integrators, with a small stepsize of $h = 0.05$ (making any qualitative sampling errors unlikely). We started the simulations with very small net momenta ($\ell_x = 0.0035$, $\ell_y \approx -0.0019$) and observed a dramatic increase in magnitude of both components when using the simple (deterministic) Berendsen scheme. If the system is ergodic in Euclidean

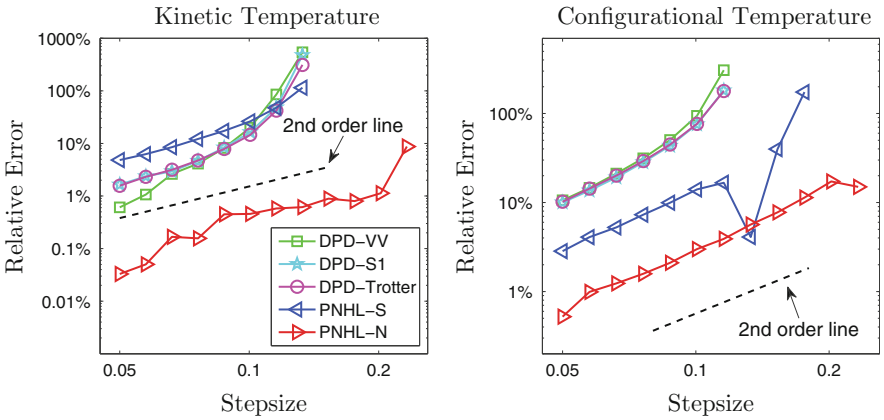


Fig. 8.9 Log-log plot of the relative error in kinetic temperature (*left*) and configurational temperature (*right*) against stepsize by using various numerical methods. The system was simulated for 1,000 reduced time units but only a trajectory's tail of time length 800 was used to calculate the static quantity to make sure the system was well equilibrated. The stepsizes tested began at $h = 0.05$ and were increased incrementally by 15 % until all methods either were above 100 % relative error or became unstable

space, the total momenta are sums of normally distributed random variables with mean zero and so the total momenta too, should have normal distributions with mean zero. Since Langevin dynamics does not respect the momentum constraint, the normal distribution is obtained. By contrast, the momentum vector is preserved exactly using the PNHL method. The NHL scheme performs somewhat oddly. In the Nosé-Hoover-Langevin method, the total momentum fluctuates but remains positive or negative if respective initial conditions are positive or negative (see the exercises at the end of the chapter). This behavior does not however contradict its mentioned ergodic properties (one needs to pay close attention to the assumptions being placed on the system in the proof of the ergodicity).

We also briefly compare DPD and PNHL methods. In Fig. 8.9 we show the kinetic and configurational temperature errors as functions of the stepsize. The results show the advantage of using the PNHL in terms of providing a more accurate sampling for given stepsize; the result is especially important since the PNHL methods are also more efficient in practice. In Fig. 8.10 the velocity autocorrelation functions are contrasted for different values of the friction coefficient. This graph shows that for small values of γ the DPD and PNHL methods perform similarly, whereas at large friction the DPD decay of correlation is substantially more rapid than that for PNHL. For some additional discussion on these points, see [228].